

Service Oriented Web Applications

Mitsiu Alejandro Carreño
Sarabia



Agenda

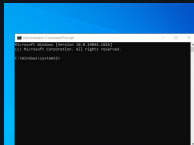
- Class requirements
- Course structure
- Functions
- Homework

Class requirements

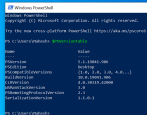
Terminal mastery

The terminal window is a powerful tool which will allow us to execute commands and programs, **you are required to learn how to use the terminal**

CMD



Powershell

A screenshot of the Windows PowerShell window. The title bar reads "Windows PowerShell". The window content shows the text: "Windows PowerShell Copyright (c) Microsoft Corporation. All rights reserved. Try the new cross-platform PowerShell https://aka.ms/powershell PS C:\Users\mehmetk> \$env:Path
Path

PathName

PathType

PSComputerVersion [1.0.1.0, 2.0.4.0...]
BuildVersion 10.0.19041.1015
CLRVersion 4.8.10115.0000
WindowsPowerShellVersion 5.0
SerializationVersion 1.1.0-1
PS C:\Users\mehmetk>

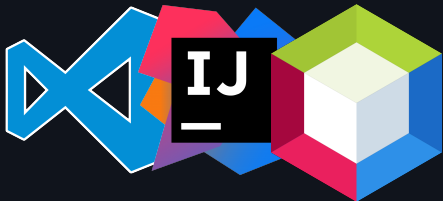
Editor mastery

Most of our time and probably a significant part of your professional time will be spent at your code editor, get to know it!

VsCode

IntelliJ

NetBeans



Most of our time and probably a significant part of your professional time will be spent at your code editor, get to know it!

- Search in a single file
- Search in multiple files
- Know filename and file path of open file
- Go to definition
- Split screen
- Go to a specific line in a file
- Find and replace in a single/multiple files

Course structure

Shared slides

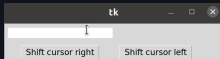
All this slides will be shared, as well as source code and extra material so here's my recommendation:

- Take hand written notes in a notebook
- Write down definitions
- Write down examples or analogies which makes sense to you
- Write down your interpretation of the lecture

My setup

I personally prefer text-based rather than graphical interfaces, it's faster!

- Mis-clicks
- Loose the cursor
- Time spent to move cursor



You can use any editor or environment which suits you, I encourage you to **learn your tools** so you can set them up for your use and comfort.

My setup

But in case of wonder here's a quick overview of my setup:

OS: MacOS (Unix based) Terminal: Kitty

(Multiplexer)

www.kovidgoyal.net



My setup

But in case of wonder here's a quick overview of my setup:

Editor: Neovim
(Fast)

<https://neovim.io/>

Slides: Presenterm (Slides as code)

<https://github.com/mfontanini/presenterm>



Everything
as Code



Course materials

Requirements:

Mandatory:

- Computer
- Master your source code editor
- Commitment
- Computers ready to share screen

Optional but recommended:

- Physical notebooks
- Get a lot of tokens
- Disable AI assistants (copilot, chatgpt, etc)
- Disable autocompletion in your editor

Course structure Partial test

- **1st Partial = 30%**
 - Autoevaluation = 5%
 - Theoretical evaluation = 40%
 - Practical evaluation (paper based?) = 55%
- **2nd Partial = 30%**
 - Autoevaluation = 5%
 - Theoretical evaluation = 40%
 - Practical evaluation (paper based?) = 55%
- **3rd Partial = 40%**
 - Autoevaluation = 5%
 - Theoretical evaluation = 40%
 - Practical evaluation (paper based?) = 55%

Wellness

I would like you to think of the classroom as



I am a guide, the effort and payback is all yours.

Wellness

I will not tolerate cheating. Evaluations are expected to be a real measurement of your knowledge.

I know this can be stressing, that's why we have tokens.

Wellness

I will giveaway **tokens** an in-class currency valued as:

1 token = 0.1 extra in your partial grade

Tokens are earned by showing interest in the class, this broad concept can be implemented as:

- Answering thoughtfull questions
- Asking thoughtfull questions
- Researching class-related topics
- Teaching/expaining to other classmates
- Specific in-class activities
- Showing mastery at activities
- (many more)

Wellness

The things you will learn in this class are important, but remember that it's just a class. If you're sleeping 4 hours at night, skipping meals or otherwise compromising your health for the sake of this class, something is wrong.

Take care of yourself if you are experiencing difficulties, don't hesitate to reach out talk to me. I am a full-time professor and can find my at my cubicle.

Let's make a plan to help you succeed.

Course intro

- From 0 (none) to 5 (a lot) how much do you enjoy writting code?
- Which is the most complex program you've written?
- Which programming languages is your favourite?
- Do you know how to use git?
- **Do you enjoy reading?** (Not limited to books)
- **Do you enjoy solving problems?** (Not limited to software)
- Did you enjoy your OOP course?
- Did you enjoy your Web Apps & Mobile Dev course?

Course intro

My class is an spiritual successor of Web Application & Mobile development class.

So show me what you achieved in it!

Diagnostic Evaluation

- Piece of paper
- Name it



Diagnostic Evaluation

1. Para qué sirve un tipo de dato
2. Ejemplifica 3 tipos de datos de cualquier lenguaje de programación
3. Para qué sirve una función
4. Ejemplifica una definición de función en cualquier lenguaje de programación
5. Para qué sirve un parámetro en una función
- 6.Cuál es la diferencia entre definir una función e invocar/aplicar una función
7. Para qué sirve una clase en el paradigma de programación orientado a objetos
- 8.Cuál es la diferencia entre definir una clase e instanciar un objeto
9. Ejemplifica definir una clase en cualquier lenguaje de programación orientado a objetos.
10. Ejemplifica instanciar un objeto en cualquier lenguaje de programación orientado a objetos.

Functions

What does the following code does?

```
function square(number) {  
  return number * number;  
}
```

What is a function?

a set of statements that performs a task or calculates a value, it should take some input and return an output where there is some obvious relationship between the input and the output.

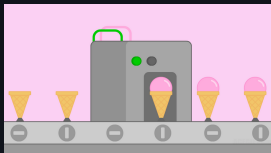
Mozilla Developer Network

Functions

What is a function?

a set of statements that performs a task or calculates a value, it should take some input and return an output where there is some obvious relationship between the input and the output.

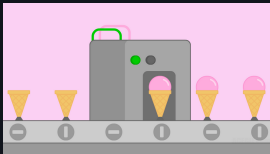
Mozilla Developer Network



Functions

What does the following code does?

```
function square(number) {  
  return number * number;  
}
```



Functions

```
function square(number) {  
  return number * number;  
}
```

A function consists of the function keyword, followed by:

- The name of the function `square`
- A list of parameters to the function, enclosed in parentheses and separated by commas `number`
- The JavaScript statements that define the function, enclosed in curly braces, { /* ... */ }. `return number * number`



Functions

```
// Help me define a function that given a number,  
outputs the number by two
```

A function consists of the function keyword, followed by:

- The name of the function **square**
- A list of parameters to the function, enclosed in parentheses and separated by commas **number**
- The JavaScript statements that define the function, enclosed in curly braces, { /* ... */ }. **return number**
*** number**



Functions

```
// Help me define a function that given an non-empty  
array returns the first element
```

A function consists of the function keyword, followed by:

- The name of the function `square`
- A list of parameters to the function, enclosed in parentheses and separated by commas `number`
- The JavaScript statements that define the function, enclosed in curly braces, `{ /* ... */ }`. `return number`
`* number`



Functions

```
// Help me define a function that tells true if a  
string is "programming"
```

A function consists of the function keyword, followed by:

- The name of the function `square`
- A list of parameters to the function, enclosed in parentheses and separated by commas `number`
- The JavaScript statements that define the function, enclosed in curly braces, `{ /* ... */ }`. `return number`
`* number`

Functions

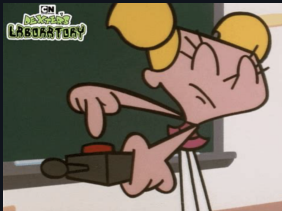
```
function square(number) {  
  return number * number;  
}
```

Defining a function is building the machine



Functions

How do we use the machine?



Notice at this point the machine is completed.

```
square(5);
```

We provide an input

30 / 47

Functions

1. We define a function (create a machine)

```
function square(number) {  
  return number * number;  
}
```

2. We invoke/apply the function

```
square(5);
```

Functions

Let's try our functions in our computer

1. Open your browser
2. Access the development tools
3. Select `console` tab
4. Define and invoke our functions

 Play/Explore

What do you do in this kind of situations?





- square function

1. What happens if we send a character?
2. What happens if we send a negative number?
3. What happens if we send a decimal number?
4. What happens if we send a really really really big number?

- byTwo

1. What happens if we send a character?
2. What happens if we send a negative number?
3. What happens if we send a decimal number?
4. What happens if we send a really really big number?
5. Does it behave differently than square?



- first

1. What happens if we send a number?
2. What happens if we send a word?
3. What happens if we send an array of booleans?
4. What happens if we send an empty array?
5. What happens if we send no parameter?

- isProgramming

1. What happens if we send "PROGRAMMING"?
2. What happens if we send "Programming"?
3. What happens if we send "programmin g"?
4. What happens if we send a number?
5. What happens if we send a boolean?

Functions and parameters

```
// Help me define a function that adds five numbers
```

A function consists of the function keyword, followed by:

- The name of the function `adder`
- A list of parameters to the function, enclosed in parentheses and separated by commas `num1, num2, num3, num4, num5`
- The JavaScript statements that define the function, enclosed in curly braces, `{ /* ... */ }`. `return num1 + num2 + num3 + num4 + num5`



- adder
1. What happens if we name all the parameters num?
 2. What happens if we only provide 4 parameters?
 3. What happens if we provide 5 strings?
 4. What happens if we provide 5 booleans?
 5. What happens if we mix 3 numbers with 2 strings?



Productive functions

```
// Help me define a function that greet people  
returning "Hola Rafael" or "Hola Diana" for any given  
name
```

A function consists of the function keyword, followed by:

- The name of the function
- A list of parameters to the function, enclosed in parentheses and separated by commas
- The JavaScript statements that define the function, enclosed in curly braces, { /* ... */ }. `return "Hola"`
`+ algunNombre`



Productive functions

```
// Help me define a function that returns if a given  
number is odd or even.
```

A function consists of the function keyword, followed by:

- The name of the function
- A list of parameters to the function, enclosed in parentheses and separated by commas
- The JavaScript statements that define the function, enclosed in curly braces, { /* ... */ }. `if (input % 2 == 0) { ... }else{ ... }`



Productive functions

```
// Help me define a function that returns the greater  
given two numbers
```

A function consists of the function keyword, followed by:

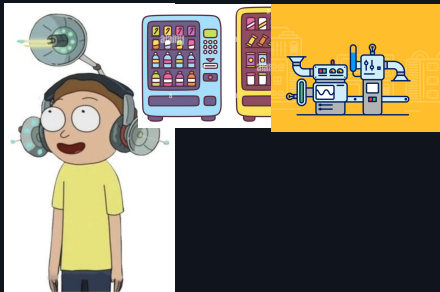
- The name of the function
- A list of parameters to the function, enclosed in parentheses and separated by commas
- The JavaScript statements that define the function, enclosed in curly braces, { /* ... */ }.

 Functions Describe the world

<https://www.youtube.com/watch?v=zHU1xH6Ogs4>

Functions Describe the world

What's the input/output?



Functions Describe the world

Let's focus on this animation:



How many machines are in this animation?

How can we translate this animation into functions?

Homework

Homework

1. Bring your computer next session

2. Master your terminal:

- `cd <path>` Change to a specific directory
- `cd ..` Go to parent directory
- `pwd` Print current directory

3. Master your code editor:

- Search in a single file
- Search in multiple files
- Know filename and file path of open file
- Go to definition
- Split screen
- Go to a specific line in a file
- Find and replace in a single/multiple files

Homework

4. Master your keyword

- How to keypress `() [] {}`
- <https://monkeytype.com/>
- Practice PascalCase with shift key

5. Upskill your english